Edge AI Lab 2 Report: Part 2 LLM Compression

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Setup

Environment. Qwen/Qwen3-4B in fp16 on one NVIDIA RTX A5000 (24 GB), Python 3.11, torch 2.5.1+cu121, transformers 4.57.1. All PPL values come from run_eval.py (WikiText-2 test, 2048-token windows). The fp16 model gets PPL 13.64.

Target layers. All nn.Linear layers in the 36 decoder blocks (q, k, v, o, gate, up and down projections), which is 252 layers and 3,633.3M weights. lm_head is tied to the embedding, so it is not compressed.

Size accounting. Asymmetric group quantization stores one fp16 scale and one b-bit zero point per group of g weights. With n_ℓ weights and B_ℓ stored bits in layer ℓ:

$$\text{bits}(b,\ g) = b + \frac{16 + b}{g}, \quad \text{CR} = 1 - \frac{\sum_\ell B_\ell}{16 \sum_\ell n_\ell}$$

SVD factors and all quantization parameters are counted in the stored bits. Pruned channels are cut out of the matrices, so no mask is stored.

Calibration. 128 random 2048-token windows from the WikiText-2 train split. From the inputs X of each layer I build the Hessian $H = \frac{2}{N} X^\top X$. Layers are compressed block by block, and each block takes its inputs from the already compressed blocks.

Bits b	Group g	Bits per weight	CR if used everywhere
4	128	4.156	74.02%
3	128	3.148	80.32%
3	256	3.074	80.79%
3	512	3.037	81.02%
8	128	8.188	48.83%

Table 1: Storage cost of group quantization.

Case 1: Compression Ratio > 75%, PPL < 15

Compression Strategy

- Method: GPTQ with g = 128. All layers use 4-bit except gate_proj and up_proj of blocks 9 to 26 (36 layers), which use 3-bit.
- Why mixed: 4-bit alone gives 74.02% (Table 1). Starting from the middle block, I switch gate_proj and up_proj to 3-bit until CR passes 75%. Attention, down_proj and the first and last blocks stay at 4-bit.
- GPTQ settings: act-order, static groups, block size 128 and damping $\lambda = 0.01 \cdot \text{mean}(\text{diag}\ H)$. Inside a block the order is q/k/v, o, gate/up, down. Each group picks its scale and zero point by an MSE search over clip ratios {1.0, 0.95, 0.9, 0.85, 0.8}.

Result

Setting	Compression Ratio	PPL
fp16	0.00%	13.64
Case 1	75.58%	13.99

Table 2: Case 1 result.

Analysis

- PPL rises by only 0.35. Three quarters of the weights stay at 4-bit, and GPTQ pushes the rounding error of each column onto the columns that are not quantized yet.
- The 3-bit layers are 24.7% of the weights and add 1.56 points of CR, which is enough to pass 75%.

Case 2: Compression Ratio ≥ 81%, PPL < 20

Compression Strategy

- Method: GPTQ 3-bit on all 252 layers with g = 512. Other settings are the same as Case 1.
- Why g = 512: 81% allows at most 0.19 × 16 = 3.04 bits per weight. g = 128 and g = 256 cost too much (Table 1), while g = 512 costs 3.037 bits. No 2-bit layers are needed.

Result

Setting	Compression Ratio	PPL
Case 2	81.02%	15.40

Table 3: Case 2 result.

Analysis

- PPL goes from 13.99 (Case 1) to 15.40. With 8 levels and one scale for 512 weights, a single outlier makes the step size larger for the whole group.
- The MSE clip search and act-order reduce this effect, so PPL stays well under 20.

Case 3: SVD Compression Ratio ≥ 20%, Compression Ratio = 50%, PPL < 50

Compression Strategy

- Method: activation-whitened SVD (as in SVD-LLM) on all 252 layers, then 8-bit round-to-nearest quantization (g = 128) of the factors of part of the layers.
- Whitening: each weight W is replaced by two Linear layers A and B, computed as

$$H = L L^\top, \quad W\ L = U\ \Sigma\ V^\top, \quad A = U_r \Sigma_r, \quad B = V_r^\top L^{-1}, \quad W \approx A\ B$$

- Cutting the rank after multiplying by L keeps the error small on the real layer inputs.
- Rank: the largest multiple of 128 with $r(m + n) \le 0.8\ m\ n$, so r = 1152 for q and o, 512 for k and v, and 1536 for gate, up and down. SVD alone removes 25.06% of the weights.
- Factor quantization: factors are set to 8-bit (8.188 bits per weight) in layer order until CR reaches 50%. This covers 174 of 252 layers (blocks 0 to 23 and the first 6 layers of block 24). The rest stay in fp16.

Result

Setting	SVD Compression Ratio	Compression Ratio	PPL
Case 3	25.06%	50.22%	23.15

Table 4: Case 3 result.

Analysis

- Lowest CR but highest PPL (+9.51 over fp16). The rank cut removes directions from every layer, while quantization only adds rounding noise.
- 8-bit quantization is usually close to lossless, so I think most of the loss comes from the rank cut. I did not run SVD without quantization to check this.
- Rounding the rank down to a multiple of 128 removes 25.06% instead of 20%. A finer step would keep more singular values.

Case 4: Pruning Rate ≥ 20%, Compression Ratio ≥ 85%, PPL < 50

Compression Strategy

- Method: structured pruning of FFN channels, then GPTQ 3-bit with g = 128 on all 252 layers. Attention is not pruned.
- Pruning: each block drops 3200 of its 9728 channels (rows of gate_proj and up_proj, columns of down_proj) with the lowest score $s_j = \|W_{:,j}\|^2\ H_{jj}$, where W is the down_proj weight and H is the Hessian of its input.
- Compensation: with K the kept and R the removed channels, down_proj is refit by least squares:

$$W_K \leftarrow W_K + W_R H_{RK} H_{KK}^{-1}$$

- Rates: $p = \dfrac{36 \cdot 3200 \cdot 3 \cdot 2560}{3{,}633.3\text{M}} = 24.35\%$ and $\text{CR} = 1 - (1 - p) \cdot \dfrac{3.148}{16} = 85.11\%$.

Result

Setting	Pruning Rate	Compression Ratio	PPL
Case 4	24.35%	85.11%	17.53

Table 5: Case 4 result.

Analysis

- Highest CR, and PPL 17.53 is still lower than Case 3 at 50%.
- Removing whole channels needs no mask, so each pruned weight saves all 16 bits. With a 1-bit mask, the same setting would only reach 78.86%.
- FFN layers hold 74% of each block, so pruning only the FFN reaches 24.35% and attention stays intact. The saved budget also allows g = 128 instead of 512.

Overall Conclusion

Setting	Method	CR	PPL	Target	Status
fp16	None	0.00%	13.64		
Case 1	GPTQ, 4-bit + 3-bit, g128	75.58%	13.99	CR > 75%, PPL < 15	Pass
Case 2	GPTQ, 3-bit, g512	81.02%	15.40	CR ≥ 81%, PPL < 20	Pass
Case 3	Whitened SVD + 8-bit factors	50.22%	23.15	SVD ≥ 20%, CR = 50%, PPL < 50	Pass
Case 4	Pruning + GPTQ 3-bit, g128	85.11%	17.53	Prune ≥ 20%, CR ≥ 85%, PPL < 50	Pass

Table 6: Summary of the four cases.

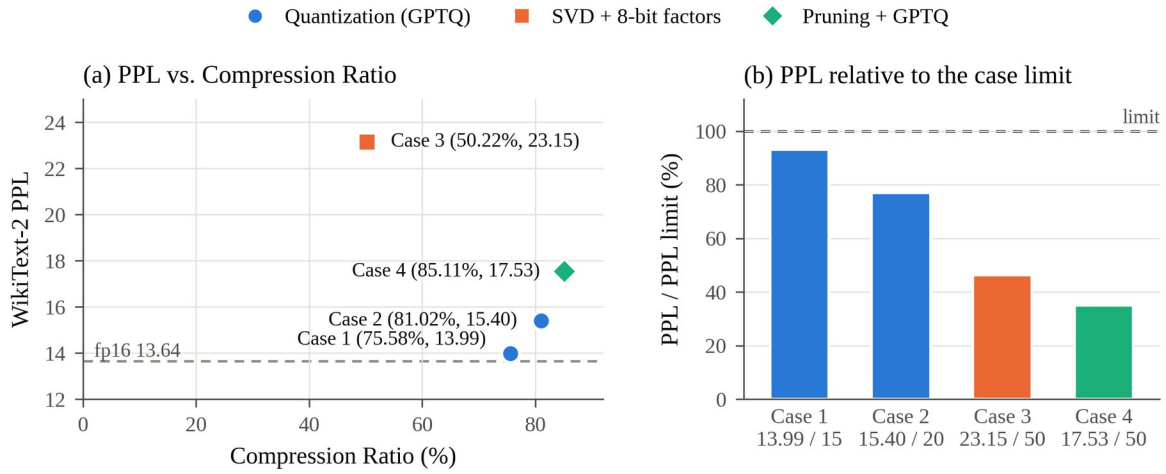


Figure 1: (a) PPL against Compression Ratio. (b) PPL of each case as a share of its PPL limit.

- Trade-off: PPL grows with CR, but the method matters more. Case 4 at 85.11% has a lower PPL than Case 3 at 50.22%.
- Sensitive layers: I did not test layers one by one, so I cannot rank them. A low-rank cut on every layer hurt the most, while FFN channels could be removed once down_proj was refit.
- Strategy differences: quantization keeps the full matrix and adds rounding noise. SVD and pruning remove parts of it. All three use the calibration Hessian to correct their error.
- Why some work better: keeping full matrices at low precision hurts less than removing parts. Case 2 gets 15.40 at 81.02%, Case 3 gets 23.15 at 50.22%.
- Limitations: the quantization is simulated, so there is no real memory or speed gain. Calibration and test both use WikiText-2, so PPL may be optimistic on other text. I did not search bits or ranks per layer, and each case was run once (seed 0).
- Summary: GPTQ gives the best PPL per stored bit, so I consider it the most effective strategy here. For CR above 85%, I would combine it with structured FFN pruning as in Case 4.